# COMMUNITY TOOL & EQUIPMENT LIBRARY

*Requirements Specification – Problem Statement #52*

## Functional Requirements

| Name | ID | Description | Priority | Acceptance Criteria |
|---|---|---|---|---|
| User Registration & Login | **FR-001** | The system shall allow community members to register and log in to access library services. | **05** | Pass: A valid member can register/login and access the system.<br>Fail: Invalid credentials are rejected. |
| Search Tools & Equipment | **FR-002** | The system shall allow members to search for power tools, camping gear, and lawnmowers. | **03** | Pass: Matching equipment and availability are displayed.<br>Fail: Search results are incorrect or unavailable items are shown as available. |
| Reserve / Checkout Tool | **FR-003** | The system shall allow an eligible member to reserve or check out available equipment for a permitted loan period. | **01** | Pass: An available tool is reserved for a valid duration.<br>Fail: A restricted checkout is allowed despite an overdue item. |
| Calculate Security Deposit | **FR-004** | The system shall calculate the required security deposit based on tool category and loan duration. | **02** | Pass: The calculated deposit matches the configured deposit rule.<br>Fail: An incorrect or missing deposit is used. |
| Return & Damage Assessment | **FR-005** | The system shall record returned equipment, assess its condition, and log any damage identified. | **04** | Pass: The return contains a condition/damage record.<br>Fail: The system completes a return without recording condition. |

## Non-Functional Requirements

| Name | ID | Type | Description | Priority | Acceptance Criteria |
|---|---|---|---|---|---|
| Real-Time Availability | **NFR-001** | Performance | **The system shall display real-time physical availability of tools across neighborhood depot lockers.** | **01** | Pass: Displayed availability matches the current depot/locker status during testing.<br>Fail: Availabi lity is stale or inconsistent. |
| Inventory Search Response | **NFR-002** | Performance | **The system shall return inventory availability search results within 2 seconds under normal operating load.** | **02** | Pass: At least 95% of test searches complete within 2 seconds.<br>Fail: More than 5% exceed 2 seconds. |

**Source:** The scenario covers resource sharing, tool availability, security deposits, loan duration, return dates, damage assessment, and late-return fees.